

Chapter 116 — Retropubic Suspension Surgery for Incontinence in Females — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Retropubic suspension needs urethral mobility** — fixed urethra / pure ISD (intrinsic sphincter deficiency) → sling or AUS (artificial urinary sphincter).
2. **Burch → Cooper's ligament**; MMK (Marshall–Marchetti–Krantz) → pubic periosteum; paravaginal → ATRP (arcus tendineus fascia pelvis); VOS (vagina-obturator shelf) → obturator fascia.
3. **Open Burch success** 85–90% at 1 yr → ~70% at 5 yr.
4. **Burch niche**: concomitant abdominal surgery, poor vaginal access, mesh avoidance.
5. **No concomitant abdominal surgery** → choose vaginal option, not retropubic.
6. **SISTER**: autologous sling > Burch success, but more morbid and obstructive.
7. **Ward–Hilton**: TVT (tension-free vaginal tape) ≈ colposuspension at 6 mo, 2 yr, 5 yr.
8. **CARE**: prophylactic Burch at sacrocolpopexy ↓ de novo SUI in continent females.
9. **Intraoperative cystoscopy mandatory** — no efflux → release sutures.
10. **Osteitis pubis**: MMK, 0.7–2.5%, 2–12 wk.

DEFINITIONS, PREVALENCE, AND COST

- **SUI (stress urinary incontinence)** = involuntary loss on effort (ICS, International Continence Society).
- **Commonest subtype**: SUI only (~26%) > MUI (mixed) (~16%) > UUI (urgency) only (~10%).
- **Any UI** >50% of US females >20 yr; only 25–45% seek care.
- ~10% of diagnosed SUI → surgery.
- **Nulliparous pure SUI**: BMI >25 and age >45 only independent risks.

GENERAL CONSIDERATIONS

- **SUI mechanisms**: hypermobility vs ISD; no test apportionments each.
- **Hammock hypothesis (DeLancey)**: backstop compression, not urethral position.
- **Enhorning**: pressure-transmission theory — basis of suspension.
- **Burch 1961**: 2–4 sutures/side to Cooper's; **Tanagho 1976**: more lateral, looser.
- **MMK 1949**; **Pereyra 1959** needle suspension; **VOS 1990**; **laparoscopic 1991**.
- **Needle suspension abandoned**: suture pull-through, vaginal tearing → lower cure.
- **Laparoscopic ≈ open** subjective cure; long-term cost-effective (QALY).
- **Slings ~90%** of US SUI operations by 2009.
- **AUS gold standard**: post-prostatectomy SUI in males.
- **AUS in females**: second-line (ICS, EAU, AUA); NICE → recurrent SUI.
- **Transvaginal AUS disfavoured**: infection, cuff erosion.

PATIENT SELECTION

- **Not indicated**: UUI without SUI. **MUI**: treats stress component only.
- **AUA/SUFU surgical options**: MUS (midurethral sling), autologous PVS (pubovaginal sling), Burch, bulking.
- **Cough test negative supine** → repeat standing, comfortably full bladder.
- **Cotton-swab (Q-tip) test**: >30° = hypermobility; no longer recommended (AUA/SUFU).
- **Minimum work-up**: PVR (post-void residual) + urinalysis.
- **UDS (urodynamics) not needed** in index patient if SUI seen on examination.
- **UDS indicated**: SUI history, not demonstrated objectively.

- **ISD**: VLPP (Valsalva leak point pressure) <60 cm H₂O or MUCP (maximal urethral closure pressure) <20 cm H₂O.
- **Voiding diary**: 1–3 days.
- **Short pad test** (20 min–2 h) confirms: >1.4 g positive.
- **24-h pad test**: 4.4–20 g mild; 21–74 g moderate; >75 g severe.
- **ICS 1-h test**: 500 mL over 15 min, or fill 150–300 mL / 50–75% capacity.
- **Burch failure predictors**: BMI >25, age >60, anticholinergics, prior SUI surgery.
- **MUS failure predictors**: BMI >30, smoking, diabetes, anticholinergics, prior surgery.
- **Most important selection factor**: patient goals; document mesh counselling.

SPECIFIC DISCUSSION REGARDING RETROPUBIC SUSPENSIONS

- **Space of Retzius**: avascular, preperitoneal; symphysis anterior, bladder posterior.
- **ATRP**: pubic symphysis → ischial spine; weakness → paravaginal defect.
- **Precluding**: no vaginal mobility; no urethral mobility (~5× failure); ISD sole cause.
- **Relative risks of failure**: urgency, obesity, menopause, prior hysterectomy/SUI surgery, advanced POP.
- **ISD settings**: recurrent SUI after surgery; neurogenic SUI.
- **Lead-pipe urethra**: immobile, rigid, fixed open → AUS territory.
- **Position**: low lithotomy / supine, legs gently abducted; Foley in.
- **Cherney incision**: rectus tendon divided transversely above pubis.
- **Santorini's plexus**: avoid midline dissection over urethra / bladder neck.
- **Burch sutures**: 0 or 1-0, SH needle; mid-urethra first; cephalad more lateral.
- **Pass medial → lateral**; don't enter vaginal lumen; distal tied first.
- **Knots above ligament**; leave ~2 fingerbreadths bladder neck–pubis.
- **MMK**: two suture pairs, full-thickness vaginal wall → periosteum.
- **VOS** = Burch + paravaginal hybrid.
- **Suture-held** (Burch, MMK) → V-angle, more obstructive; **scar-held** (paravaginal, VOS) → flatter, less.
- **AUS in females**: cuff at bladder neck; pump labia majora; reservoir preperitoneal.

- **Transvaginal AUS:** inverted-U incision; copious antibiotic irrigation.
- **Postoperative voiding:** CIC preferred to suprapubic tube; continue until PVR <150 mL.

OUTCOMES

- **Objective success:** cough stress test + 1-h pad test.
- **In-person review by 6 mo:** history, examination, PVR.
- **Follow-up:** short 1–2 yr; intermediate to 47 mo; long >48 mo.
- **SISTER (655 randomised):** PVS > Burch at 2 yr; more urinary tract infection, de novo UUI.
- **SISTER 5 yr:** Burch continence lower; sling satisfaction higher.
- **Ward–Hilton:** 179 TVT vs 169 colposuspension; 6-mo dry 36% vs 28%.
- **TVT:** more bladder perforation; **colposuspension:** more voiding dysfunction.
- **Retropubic MUS vs Burch (meta-analysis):** MUS higher cure, more perforation.
- **Laparoscopic Burch vs TVT:** TVT better, shorter operation.
- **Transobturator MUS vs Burch:** comparable; less retention with transobturator.
- **Surgery vs collagen:** surgery more successful; collagen fewer complications.
- **CARE (322):** Burch in stress-continent sacrocolpopexy → less SUI at 2 yr.
- **Preoperative SUI at sacrocolpopexy:** MUS > Burch.
- **POP after colposuspension:** 2.5–27%.

- **AUS:** erosion rises beyond 5 yr; minimally invasive → less mechanical failure.
- **No efflux on cystoscopy** → release sutures, repeat until patent.
- **Osteitis pubis:** suprapubic pain to legs, tender symphysis → antibiotics → debridement.

CLASSIC TRAPS

- **Trap:** Burch for fixed-urethra ISD — **Actually:** needs mobility; sling or AUS.
- **Trap:** Burch as default standalone — **Actually:** only with abdominal surgery / mesh avoidance.
- **Trap:** Burch anchors to pubic periosteum — **Actually:** that is MMK; Burch → Cooper's.
- **Trap:** Burch offered for UUI — **Actually:** not indicated without SUI.
- **Trap:** UDS routine before SUI surgery — **Actually:** not if SUI demonstrated on examination.
- **Trap:** VLPP / MUCP predicts success — **Actually:** leak point pressure does not.
- **Trap:** Burch equals sling in SISTER — **Actually:** PVS more successful, more morbid.
- **Trap:** sutures near bladder neck, midline — **Actually:** lateral; else obstruction, bleeding.
- **Trap:** osteitis pubis after Burch — **Actually:** classically MMK.
- **Trap:** transvaginal AUS preferred in females — **Actually:** open abdominal; infection, erosion.